

Predicting a latent population summary from survey estimates: what the interval is sensitive to, and what that costs

Abstract

A survey covering many regions produces one estimate for each. Those estimates are then used to describe a place the survey did not cover: a region newly added to a frame, or one too small to sample. That means separating real variation between places from sampling error, and the sampling error is itself estimated, from designs supplying few degrees of freedom. We ask what that costs.

The answer is not about estimation accuracy. Competing methods can be given exactly the same information about the unknown spread of the true values and still report very different widths, because they differ in how strongly the reported width reacts to being wrong about that spread. We define that reaction, show it is weaker for a method that subtracts the sampling error than for methods whose width is proportional to the square root of a variance bound, and prove an exact identity relating the two. Three things follow. The number of places a practitioner needs is far smaller than a requirement stated on the variance alone implies, and this is a property of the problem, not of the estimator. Two methods can change places, one narrower in one survey and wider in another, at a computable point. And in one matched case the width ratio depends on nothing but the number of places and the error budget; we recorded that prediction before testing it, and one of its three parts failed.

We then give an interval exploiting the weaker reaction, with a guarantee stated under

a normal model and a bound computed rather than searched for, and report where it stops working. On three European Social Survey questions across three rounds it is narrower than a comparable method in 44 of 45 configurations, under stated approximations and with no claim about coverage.

Statement of Significance

Survey programmes are regularly asked about places they did not reach. The usual route borrows strength from the places it did reach, which requires knowing how much of the observed variation between places is real and how much is sampling noise. That split must be estimated, often from small samples, and practitioners have little guidance on when the resulting intervals can be trusted.

This paper shows that the question usually asked—how precisely must we know the variation between places?—is the wrong one. The interval depends on the *ratio* of the two variances, and a requirement stated on the variation alone can overstate the number of places needed many times over. We also show why two similar methods can swap places, one narrower in one survey and wider in another, and identify what decides it.

The practical products are an interval computed rather than searched for, with a guarantee stated under explicit conditions; an account of where it stops working; and a diagnostic an analyst can compute from data already in hand. We are equally explicit about what is not established, above all the gap between the model in which the guarantee is proved and the complex designs of real surveys.

Keywords: small area estimation; Fay–Herriot model; design variance; variance components; intraclass correlation; conformal prediction; European Social Survey.

1 Introduction

A survey programme covering many countries, regions or domains produces one estimate of some summary per population per round. A second analysis then uses those estimates to say something about a population the survey did not cover: a region added to a sampling frame, a country joining a programme, a domain too small to have been sampled. Call the target θ_{K+1} , the latent summary of that population, and the observed inputs $Y_1, \dots, Y_K$, the survey estimates for the populations that were covered.

The obstacle is that each Y_c carries sampling error. The spread of the Y_c is larger than the spread of the θ_c , so an interval calibrated on the observed spread is too wide for the latent target. Removing the difference means subtracting a design variance from a between-population variance, and the design variances are themselves estimated—from stratified cluster designs with, in the European Social Survey regions we analyse, a median of 22 to 29 degrees of freedom per population.

Two literatures meet here and neither answers the question. Small-area estimation treats the sampling variances as known and targets areas that were sampled (Fay and Herriot, 1979; Rao and Molina, 2015); where it treats them as estimated it corrects the mean squared error of an *in-sample* predictor (Wang and Fuller, 2003; Rivest and Vandal, 2003; You and Chapman, 2006; Maiti, Ren, and Sinha, 2014; Sugasawa, Tamae, and Kubokawa, 2017). Recent conformal work reaches valid confidence intervals for in-sample areas without estimating the sampling variances at all: Zhang and Tuoto (2026) target “the unknown expectations of the in-sample outcomes . . . conditional on the realised sample”, and their split conformal construction is, in their words, “easier to implement than the direct or EBLUP interval, because there is no need at all to estimate the sampling variances” (their §3.4). That is an argument for their target, not against ours: an interval for a population the survey never covered cannot avoid separating the two scales. Conformal prediction of a future *unit* within an area (Bersson and Hoff, 2024) is a third target again. Prediction for an out-of-sample area is standard through synthetic estimation, and the linear mixed-model literature does supply prediction intervals for targets containing a new random

effect (Chatterjee, Lahiri, and Li, 2008)—but with the sampling variances treated as known.

What this paper adds. Not another way to smooth design variances, and not a claim that con-formal prediction is robust here; our exact inference rests on a normal- χ^2 model and we say so. The contribution is in two parts.

A result about the problem. Section 3 isolates the scale elasticity and shows that it, rather than the quality of the scale estimate, decides which construction is narrowest. The consequences are checkable and we check them: the number of populations required is smaller by ρ^{-4} than a criterion on the variance component implies; two constructions cross, at a computable point; and for one matched pair the scale bound cancels entirely, giving a width ratio that depends only on K and the error budget. That last prediction was registered before the experiment that tests it, and one of its three registered claims was refuted—the constant, by an amount we locate.

A procedure that exploits it. Section 4 constructs an interval whose sensitivity to the unknown scale is the damped one, carries the uncertainty of an estimated variance structure rather than substituting a fitted value, and is computable by a conservative bound rather than a search.

What we do not claim. The guarantee is proved in a restricted normal model. Under a complex design its conditions fail, measurably, and we report where the procedure nonetheless behaves and where it does not. On real survey data we report widths, inputs and diagnostics, and no coverage: the latent target is unobserved and a held-out population’s direct estimate is not a substitute for it.

2 Target, information, and what has to be estimated

2.1 The two objects

Write θ_c for population c ’s latent summary and Y_c for its survey estimate, taken as a deviation from a fixed centre. The observed and the latent object are not interchangeable, and the paper

keeps them apart throughout:

$$Y_c = \theta_c + e_c, \quad \text{Var}(\theta_c) = A, \quad \text{Var}(e_c) = D_c, \quad (1)$$

with D_c the design variance of the survey estimate. The *design share*

$$\rho_c^2 = \frac{D_c}{A + D_c} \in [0, 1) \quad (2)$$

is the fraction of the observed dispersion that is sampling error. An interval calibrated on the Y_c covers a further *survey estimate*; covering a further *latent summary* requires removing that fraction.

2.2 The interval

Fix an integer $m \leq K$, write ord_m for the m th smallest of K numbers, and let $t = D/A$ be the scale ratio, taken common across populations for this section. Standardising $Z_c(t) = Y_c/\sqrt{1+t}$, the interval is

$$B(t) = [-R(t), R(t)], \quad R(t) = \text{ord}_m\{|Z_c(t)|\}_{c=1}^K = q \kappa, \quad (3)$$

where $q = \text{ord}_m\{|Y_c|\}$ and $\kappa = (1+t)^{-1/2} = \sqrt{A/(A+D)} = \sqrt{1-\rho^2}$. Here q is a statistic—the radius the observed spread itself supplies—and κ is the factor carrying it to the latent scale.

Lemma 1 (Base validity). *If $\theta_{K+1}, Z_1(t), \dots, Z_K(t)$ are exchangeable with a continuous distribution, then $\Pr\{\theta_{K+1} \in B(t)\} = m/(K+1)$. Under (1) with equal design variances this holds exactly at the true t , because $Z_c(t)$ and θ_{K+1} are then i.i.d. $N(0, A)$. Write $\alpha_0 = 1 - m/(K+1)$.*

Everything that follows concerns one difficulty: t is unknown, so the interval is evaluated at a bound for it rather than at its value.

2.3 What has to be known precisely, and it is not A

The band depends on the two scales only through their ratio κ , and not on A separately. Since $\widehat{A}$ and $\widehat{A + D}$ are built from the same observed spread, their errors partly cancel, and to first order

$$\frac{\text{RSE}(\widehat{\kappa})}{\text{RSE}(\widehat{\sqrt{A}})} = \sqrt{\frac{\rho^4 \sigma_T^2 + \sigma_D^2}{\sigma_T^2 + \sigma_D^2}} \longrightarrow \rho^2 \quad (4)$$

as the design variances become well determined, where σ_T^2 and σ_D^2 are the variances of the estimated total and design scales. A requirement stated on $\widehat{A}$ therefore overstates what the band needs by a factor ρ^{-4} in the number of populations. Simulation puts the realised gain at 14 to 24 times at $\rho^2 = 0.2$ and 2.0 to 3.8 times at $\rho^2 = 0.5$, against the predicted ρ^{-4} of 25 and 4.

2.4 Two axes, not one

Writing $\sigma_D^2 = 2D^2/\nu_{\text{tot}}$ for a design-variance estimate on ν_{tot} degrees of freedom, the requirement $\text{RSE}(\widehat{\kappa}) \leq \varepsilon$ becomes

$$\boxed{\frac{1}{K-1} + \frac{1}{\nu_{\text{tot}}} \leq \frac{2\varepsilon^2(1-\rho^2)^2}{\rho^4}} \quad (5)$$

Populations and design degrees of freedom enter *harmonically*: neither substitutes for the other, and ν_{tot} imposes a floor no number of populations clears. Replicate count does not help—with design degrees of freedom $\nu^{\text{des}} = 20$, raising a resampling count from 100 to 10,000 buys 3.6 effective degrees of freedom and no more.

The boundary is not an artefact of this estimator. Condition (5) is sufficient, and it was derived from one estimator's relative standard error. A matching necessary condition holds. Restrict to equal design variances and take two parameter points separated so that their ε -balls in κ are disjoint. The scaled- χ^2 and normal Hellinger affinities are available in closed form, so Le Cam's two-point bound (Le Cam, 1986) gives, exactly and in finite samples, that *no* procedure can cer-

tify κ to tolerance ε with failure probability $\eta < 1/2$ unless

$$v_{\text{tot}} \geq L(\eta) \frac{\rho^4}{\varepsilon^2(1-\rho^2)^2}, \quad L(\eta) = -\log \{2\sqrt{\eta(1-\eta)}\}, \quad (6)$$

and the same bound with K in place of v_{tot} . The dependence on ρ and on ε is the same as in (5); the constants differ by a factor 2.4 to 2.9 over the range we checked ($\rho^2 \geq 0.4$, $\varepsilon \leq 0.05$, $\eta = 0.05$), one side being exact and the other a normal approximation. At small design share the perturbation approaches the edge of the parameter space and the bound loosens, to a factor 14 at $\rho^2 = 0.2$, $\varepsilon = 0.05$.

The two axes carry the *same* constant, and not by coincidence. Writing τ for the ratio of the two κ values, the v -axis perturbation multiplies D by $(1 - \tau^2 \kappa^2)/(1 - \kappa^2)$ while the K -axis perturbation multiplies $A + D$ by exactly its reciprocal, and the affinity $\{2\sqrt{r}/(1+r)\}^{n/2}$ is invariant under $r \mapsto 1/r$. That is why the two enter harmonically rather than in some other combination. *Neither the technique nor, in this submodel, the problem is new.* With equal design variances the model is the balanced one-way random-effects model, κ is a monotone function of the intraclass correlation, exact intervals for that coefficient are classical, and trading the number of groups against the number of observations per group to control interval length is an established design literature (Donner, 1986; Zou, 2012; Mondal et al., 2024). The minimax theory for signal-to-noise ratios in high-dimensional regression (Verzelen and Gassiat, 2018) concerns a different observation structure: there is no separate channel carrying the error scale on a known number of degrees of freedom, which is what produces the second axis here. We state (6) to establish that (5) reflects the problem rather than our estimator, and we claim nothing further for it.

2.5 Estimating the design variances costs little; the estimator costs a lot

Treating $D_1, \dots, D_K$ as unknown nuisance parameters observed through $v_c \widehat{D}_c / D_c \sim \chi_{v_c}^2$, the effective information for A is

$$I_{\text{eff}}(A) = \frac{1}{2} \sum_c \{ (A + D_c)^2 + D_c^2 / v_c \}^{-1}, \quad (7)$$

so the standard error for A inflates, relative to known design variances, by at most $\sqrt{1 + \rho^4 / v_{\min}}$: 3.2% at $v = 10$ and $\rho^2 = 0.8$, 7.7% at $v = 4$, 14.9% at $v = 2$. **At the degrees of freedom we examine, not knowing the design variances is a second-order cost.** The bound is attained exactly, in finite samples, by a moment estimator with weights $w_c \propto \{ (A + D_c)^2 + D_c^2 / v_c \}^{-1}$.

The first-order cost is the estimator. Under dispersed design variances the unweighted subtraction $\widehat{A} = K^{-1} \sum_c (Y_c^2 - \widehat{D}_c)$ has relative efficiency 23% against (7), where oracle weighting reaches 99%. And substituting $\widehat{D}_c$ into a known-variance likelihood biases $\widehat{A}$ *upward* by approximately $4\rho^4 / \{v(1 - \rho^2)\}$ relative to A , a bias of order $1/v$ that does not average away as populations accumulate. The second-order mean-squared-error corrections of Prasad and Rao (1990) and Datta and Lahiri (2000) address the same estimated-variance effect for an in-sample predictor, where the target makes the correction additive rather than a multiplicative distortion of a band.

These are properties of the estimation problem. What matters for the interval is the next section.

3 Scale sensitivity

No construction evaluates its half-width at the truth. Each evaluates it at a bound. **What separates constructions is therefore not how well they estimate the scale—they can be given the same estimate—but how hard the interval reacts to being wrong about it.**

Definition 1. For a half-width R depending on a scale θ , the *scale elasticity* is

$$e(\theta) = \frac{\partial \log R}{\partial \log \theta}.$$

What is held fixed. The derivative is taken in the scale at which the half-width is *evaluated*, with the data—and hence the empirical radius $q = \text{ord}_m |Y_c|$ of (3)—held fixed. It is not the derivative of the oracle width: at the truth every construction here has half-width proportional to $\sqrt{A}$ and so oracle elasticity $1/2$. The distinction is the substance of this section. Constructions can be given the same information about the scale and still differ in how an error in the plugged-in bound propagates into the number they report, and it is that propagation, not the quality of the estimate, that decides which is narrower.

With latent scale A and design variance L , so $\rho^2 = L/(A + L)$:

construction	half-width	elasticity in A
deconvolution band	$q\sqrt{A/(A+L)}$	$L/\{2(A+L)\} = \rho^2/2$
any band $\propto \sqrt{A}$	$z\sqrt{A}$	$1/2$
uncorrected anchor	q	0

The deconvolution band’s elasticity is **damped by the design share**. This is not a property of conformal prediction: it follows from dividing by $\sqrt{A+L}$ and multiplying by $\sqrt{A}$, and any construction of that form inherits it.

Figure 1 shows both halves of the claim: the relative cost of a loose bound on the left, and on the right the elasticity measured in eight simulation cells against the predicted $\rho^2/2$. The measurement sits below the prediction in every cell, averaging 0.86 of it—the damping is real and is slightly stronger than the first-order calculation says, which is the safe direction for a claim of the form “this construction reacts less”.

Proposition 1 (Exchange identity). *For the deconvolution band R_c and any band $R_g \propto \sqrt{A}$, evaluated at a common upper bound $A_U \geq A$,*

$$\frac{R_c(A_U)/R_g(A_U)}{R_c(A)/R_g(A)} = \sqrt{\frac{A+L}{A_U+L}} =: \Lambda.$$

Proof. $R_c(A_U)/R_c(A) = \sqrt{A_U(A+L)/\{A(A_U+L)\}}$ and $R_g(A_U)/R_g(A) = \sqrt{A_U/A}$; divide. $\square$

The identity is exact, not asymptotic, and we verified it numerically on 2×10^5 random (A, L, A_U) with no violations. Its content: $\Lambda \rightarrow 1$ as $L \rightarrow 0$ —with no design noise there is nothing to decon-

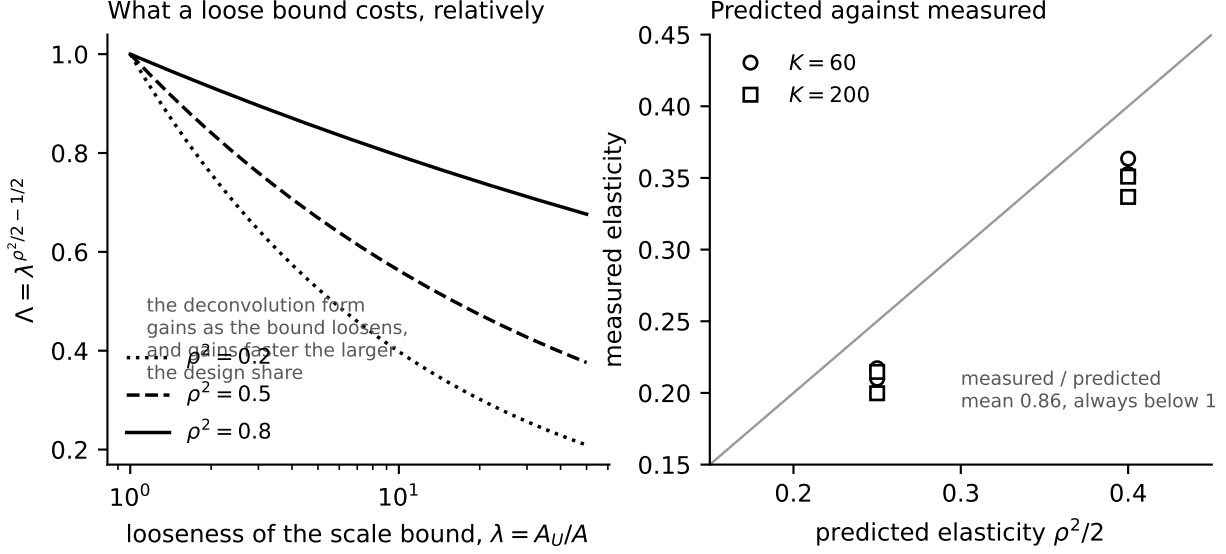


Figure 1: **Left:** the exchange identity of Proposition 1. Evaluated at a bound loose by λ , the deconvolution band's half-width moves by $\lambda^{\rho^2/2}$ where a band proportional to $\sqrt{A}$ moves by $\lambda^{1/2}$; the curves are their exact ratio, with no fitted constant. **Right:** elasticity measured in eight cells ($K \in \{60, 200\}$, $\bar{D}/A \in \{1, 4\}$, $v = 10$) against the predicted $\rho^2/2$. The line is equality.

Alt text: Two panels. Left, three descending curves on a logarithmic horizontal axis showing the relative width cost of a loose scale bound; the curve for the largest design share descends most slowly. Right, a scatter of eight points of measured against predicted elasticity, all lying below the line of equality.

volve and no advantage—and $\Lambda \rightarrow \sqrt{A/A_U}$ as $L/A \rightarrow \infty$. **The deconvolution form's advantage is bought entirely by design noise, and it is largest exactly when the scale bound is loosest.**

Consequence 1: the requirement. Certifying $\hat{A}$ and certifying the band's own scale factor to the same tolerance differ by the elasticity ratio, and the population requirement by its square, ρ^4 —the saving reported in Section 2, where (6) shows it is a property of the problem and not of the estimator used to realise it.

Consequence 2: constructions cross. Two forms with elasticities $e_1 < e_2$ and oracle width ratio r_0 realise $r_0 \lambda^{e_1 - e_2}$ at a bound loose by $\lambda = A_U/A$, and cross at $\lambda^* = r_0^{-1/(e_2 - e_1)}$: below it the higher-elasticity form is narrower, above it the lower one is. A single pair of constructions therefore changes places between survey settings. Two of our own experiments straddle that

point—one finds the normal-theory band narrower by 2 to 22%, the other finds it wider by 7 to 14%—and the elasticity reconciles them, since the first uses a tight scale limit and the second a loose one. **We do not present this as a measured crossing.** The two runs differ in more than the tightness of the limit and the later supersedes the earlier, so what the calculus supplies here is an explanation for a disagreement we had previously recorded as an error, not an experiment designed to locate λ^* .

Consequence 3: sometimes the bound cancels. If the comparator’s bound is built from the deconvolution band’s own statistic— $A_U = \bar{S}/\chi_{K,\alpha_2}^2$ with $\bar{S} = \sum_c Y_c^2/(1+\underline{t})$, against $\text{ord}_m |Y_c|/\sqrt{1+\underline{t}}$ —then $\underline{t}$, A , L , ρ and v all cancel:

$$\frac{R_{\text{deconv}}}{R_{\text{normal}}} \approx \frac{z_p}{z_{1-\alpha_1/2}} \sqrt{\frac{\chi_{K,\alpha_2}^2}{K}}, \quad p = \frac{m}{K+1}. \quad (8)$$

3.1 A registered test, and what it refuted

Equation (8) makes a strong claim: a ratio built entirely out of design variances should not depend on the design share. We wrote the predicted values into a protocol before running a fresh grid ($K \in \{40, 120, 400\}$, $\rho^2 \in \{0.20, 0.45, 0.70\}$, $v \in \{6, 20\}$, correct and misspecified structure; none of these values used in the experiment that suggested the law).

registered prediction	result	tolerance	verdict
budget-split dependence	identical to 10^{-6} , six splits	monotone, 5%	passes
level of the ratio	worst cell 7.07%	5%	fails
invariance in ρ^2, v	within- K spread 0.0200/0.0110/0.0089	0.02	marginal

The first row is the sharpest test and it passes exactly: across six splits spanning a factor of 1.32 in the predicted ratio, the discrepancy is one constant. The second row locates that constant— $c(K) = 1 + 1.9/K$, the order $1/K$ gap between an order statistic’s expectation and the corresponding quantile—and the refined form errs by at most 2.2%. The third row shows the invariance is *asymptotic*: a residual dependence on ρ^2 of 1.4% at $K = 40$ falls to 0.45% at $K = 400$. **The**

strong form of the invariance claim is refuted, and what stands is a dependence an order of magnitude smaller than the construction of the quantity would suggest.

Why this widens the comparison. The elasticity is a property of the *form* $c\sqrt{\theta}$, not of a particular baseline. Any interval whose half-width is proportional to the square root of a scale bound—a normal-theory interval, an EBLUP prediction interval for a target containing a new random effect, a parametric bootstrap interval built on one—has elasticity $1/2$ and so pays the full $\lambda^{1/2}$ for scale uncertainty. Proposition 1 therefore compares against a class rather than against a single constructed competitor. **It does not rank them absolutely**: the oracle ratio r_0 differs between constructions and is not supplied by the elasticity. What the identity gives is the differential response to scale uncertainty, which is the part that reverses.

4 A procedure that exploits the damping

Section 3 says the interval should depend on the unknown scale as weakly as it can while still covering. This section builds one that does, and makes it an algorithm.

4.1 Model

Assumption 1 (Model R). Independently across $c = 1, \dots, K$, with μ known,

$$Y_c \sim N(0, A + d a_c), \quad \frac{v_c \hat{D}_c}{d a_c} \sim \chi_{v_c}^2, \quad Y \perp \hat{D},$$

where $a_c > 0$ is a *relative* design-variance structure, $d > 0$ an unknown common scale, and $A > 0$ the between-population variance. Write $t = d/A$, so that $D_c/A = a_c t$: one unknown ratio parameter.

Assumption 1 is the heteroscedastic model in which reduction to one ratio parameter is *true* rather than assumed. A general heteroscedastic model does not reduce this way, and substituting $\hat{D}_c$ for D_c to force it would reintroduce exactly the fault the construction exists to avoid.

4.2 An exact pivot for the correction ratio

Proposition 2. *Under Assumption 1, with $\mathbf{v} = \sum_c \mathbf{v}_c$, $\hat{d} = \mathbf{v}^{-1} \sum_c \mathbf{v}_c \hat{D}_c / a_c$ and $S(t) = \sum_c Y_c^2 / (1 + ta_c)$,*

$$T(t) = \frac{K\hat{d}}{tS(t)} \sim F_{\mathbf{v},K} \quad \text{exactly,}$$

and $\partial_t \{tS(t)\} = \sum_c Y_c^2 / (1 + ta_c)^2 > 0$, so T is strictly decreasing and inverts to an exact one-sided lower bound t_L , the solution of $T(t_L) = f_{\eta_t}$ with f_{η_t} the upper η_t quantile of $F_{\mathbf{v},K}$, satisfying $\Pr\{t \geq t_L\} = 1 - \eta_t$.

The band is (3) with the heteroscedastic standardisation, so its oracle half-width is

$$R(t) = \text{ord}_m \{ |Y_c| / \sqrt{1 + a_c t} \},$$

and Lemma 1 applies at the true t with $\alpha_0 = 1 - m/(K+1)$. Since $R(t)$ is decreasing in t , $R(t_L)$ bounds it above on that event. **One parameter, one budget, one bound**—against a construction that spends one budget on an upper limit for A and another on K simultaneous lower limits for the D_c , both of which drive the correction toward zero.

4.3 An estimated structure, paid for rather than substituted

With $a_c(\gamma) = x_c^{-\gamma} / \overline{x^{-\gamma}}$ for a design variable x_c known before any estimate is seen, write $\varepsilon_c = \log(\chi_{\mathbf{v}_c}^2 / \mathbf{v}_c)$, whose law is known. The statistic

$$W(\gamma) = \sum_c (\log x_c - \overline{\log x}) \{ \log \hat{D}_c + \gamma \log x_c \}$$

has, at the true γ , a null law depending on neither d nor γ , and is linear increasing in γ . Hence

$$C_\gamma = \left[\hat{\gamma} - \frac{w}{S_{xx}}, \hat{\gamma} + \frac{w}{S_{xx}} \right], \quad \hat{\gamma} = -S_{xD} / S_{xx}, \quad (9)$$

in closed form, with $\Pr\{\gamma_0 \in C_\gamma\} \geq 1 - \eta_\gamma$. The interval is then

$$R_U = \sup_{\gamma \in C_\gamma} R(t_L(\gamma), \gamma),$$

and on $\{\gamma_0 \in C_\gamma\} \cap \{t \geq t_L(\gamma_0)\}$ it dominates the oracle half-width, so

$$\Pr\{\theta_{K+1} \notin B_U\} \leq \alpha_0 + \eta_\gamma + \eta_t. \quad (10)$$

No independence between the two confidence events is required.

4.4 Making the supremum a computation

A grid maximum is not a supremum, and **we found by counterexample that the minimiser of $t_L(\gamma)$ can be interior**, 16.9% below the smaller endpoint, so an endpoint evaluation is not available. A bound is.

Lemma 2. $\partial_\gamma \log a_c(\gamma) = \bar{\ell}(\gamma) - \log x_c$ with $\bar{\ell}'(\gamma) = -\text{Var}_w(\log x) \leq 0$, so $\log a_c$ is concave in γ and $\inf_{\gamma \in [\gamma_L, \gamma_U]} a_c(\gamma) = \min\{a_c(\gamma_L), a_c(\gamma_U)\}$ exactly.

Proposition 3 (Certified bound). *Let $\underline{a}_c$ be that endpoint minimum and $\bar{\Phi}(t) = \sum_c t Y_c^2 / (1 + t \underline{a}_c) \geq t S(t; \gamma)$ for every $\gamma \in C_\gamma$. Since $1/a_c(\gamma) = K^{-1} \sum_j (x_c/x_j)^\gamma$ is a positive sum of exponentials, $\hat{d}(\gamma)$ is convex, and a bisection on its increasing derivative brackets the minimiser, giving a certified $\underline{d}$. Then*

$$\underline{t} = \bar{\Phi}^{-1}(K \underline{d} / f_{\eta_t}) \leq \inf_{\gamma \in C_\gamma} t_L(\gamma),$$

and $\text{ord}_m\{|Y_c|/\sqrt{1 + t \underline{a}_c}\} \geq R(t_L(\gamma), \gamma)$ for every γ in the set.

The bound needs two structure evaluations, one convex minimisation and one monotone inversion, which is cheaper than the grid it replaces. An optimiser's returned value is not a lower bound and is not used: the endpoint cases are exact and the interior case uses a supporting line at a certified bracket, with every returned quantity shrunk by a relative 10^{-9} for rounding. Verified: no violations in 400 stress configurations, tightness median 0.941.

Unknown centre. With $Y \sim N(X\beta, AI + \text{diag}(\mathbf{D}))$ and $\text{rank}(X) = p$, refitting the weighted regression at each candidate gives $Q(A, \mathbf{D}) = \min_{\beta} \sum_c (Y_c - x'_c \beta)^2 / (A + D_c) \sim \chi^2_{K-p}$, and the argument extends. **This bounds the variance component only:** it says nothing about the error from centring the band on an estimated mean, or about exchangeability of the scores under an estimated centre. Both matter in Section 6.

5 Simulation: what holds, and where it stops

Three questions, and the third is the one that decides whether the procedure is worth using.

5.1 Against a comparator sharing the target, information and level

The direct comparator is a normal-theory prediction interval for the *same* new population’s latent value, built from the *same* structure confidence set and the *same* certified scale bound, spending the identical budget $\alpha_0 + \eta_\gamma + \eta_t = 0.10$: $A_U = \bar{S} / \chi^2_{K, \alpha_2}$, the exact chi-square limit rather than a Satterthwaite or Graybill and Wang (1980) approximation (Satterthwaite, 1946), and $\pm z_{1-\alpha_1/2} \sqrt{A_U}$ with $\alpha_1 + \alpha_2 = \alpha_0$. The two arms differ in exactly one place.

K	proposed cov.	comparator cov.	proposed narrower by	split
60	0.957–0.978	0.982–0.996	22.2%	scanned in its own favour
60	—	—	25.9%	fixed in advance
200	0.957–0.970	0.983–0.988	13.9%	scanned in its own favour
200	—	—	20.4%	fixed in advance

All against a guaranteed 0.90. The pre-registered split is the primary comparison; the scanned one gives the comparator its best budget allocation and is the smaller number, so **we quote 13.9 to 22.2% throughout**. Two readings follow. Both arms over-cover, but the proposal is *both* narrower and closer to the nominal level, at 0.957–0.978 against 0.982–0.996: the width difference is not bought by spending coverage. And the comparator is one valid construction, not a class and not the best possible—the registered prediction before this experiment was that it would win, and it did not.

A disagreement between two of our own runs. On a tight scale limit the normal-theory band is narrower by 2 to 22% in eight cells; on a loose pivot limit an earlier run found it wider by 7 to 14%. We recorded the second as an error and superseded it. Consequence 2 of Section 3 says the two are the two sides of λ^* and neither need be wrong. We report this as a reconciliation and not as evidence: the runs differ in more than the tightness of the limit, and an experiment that varies only λ and locates the crossing is one we have not done.

5.2 The regimes where it stops

Under finite-population stratified cluster sampling with a pre-specified sample-size structure variable, the picture divides not by design share but by what else changes with it.

	degrees of freedom 12–28	degrees of freedom 4–8
structure R^2	0.59–0.85	0.05–0.71
zero variance estimates	none	up to 17%
containment	0.955–0.992	0.681–0.987
latent coverage	0.958–0.971	0.944–0.975
band vs. structure-free	0.883–0.937	0.836–0.928

At matched $\bar{D}/A \in \{0.55, 0.75\}$ the two columns differ in degrees of freedom, in the rate of exactly-zero variance estimates and in the fit of the structure model—**the same noise ratio is not the same inference problem**. In the first column the procedure holds its containment requirement and is 6.3 to 11.7% narrower than a structure-free construction, capturing 0.56–0.74 of the available narrowing against 0.23–0.30. In the second it loses containment at a tail quantile and its width figures are not usable.

Supplying the *true* structure raises containment there from 0.68 to 0.82 and from 0.77 to 0.89 and does not reach 0.95. Two conclusions and only two: approximating the variance structure affects the outcome, and improving that step alone would not repair the procedure. We do not apportion the residue among normality, the χ^2 law and independence; this design breaks all three at once and the experiment does not separate them.

We report the verified scenario range, not a usage threshold. Writing “degrees of freedom above 12 and structure R^2 above 0.6” as an operating rule would be wrong three times: those numbers co-occurred in these scenarios rather than being derived as thresholds; the structure R^2 is computed from the *true* design variances and is unavailable in real data; and nothing here proves normality or the χ^2 law under any complex design.

5.3 Honest accounting of the gain

At the regional design share the *valid* interval is 0.9 to 6.6% narrower than not correcting at all, capturing 6 to 31% of what the oracle achieves. The reason is the guarantee itself: the argument needs A bounded above and each D_c bounded below, and both drive the correction toward zero. Measuring where the width goes shows the simultaneity correction is only 6 to 22% of that loss, so replacing it alone would not repair matters—which is why the construction bounds the ratio rather than the components.

6 European Social Survey

6.1 What the data can and cannot show

The latent θ_{K+1} is unobserved. **No coverage is claimed or computed here**, and a held-out region’s direct estimate is not a substitute for the latent value: treating it as one would measure the uncorrected anchor. What real data can show is which inputs each construction needs, how far the intervals differ on the same data, and which assumptions the data support.

6.2 Design

Rounds 9–11, 127,286 respondents, 75 country-rounds, 825 region-rounds, retrieved through the portal API so that the analysis reproduces from source for any user with an End User Licence. Everything was fixed before any estimate was computed: the target is a single CDF point $F_{cr}(t_0)$

with t_0 set by rule (one below the scale midpoint); three items chosen in advance for differing between-region heterogeneity; the structure variable x_{cr} is the region’s sampled PSU count, carried over unchanged from the simulations; one budget for every arm; the comparator’s split fixed rather than scanned. Regions with fewer than two design degrees of freedom (18 of 825) are excluded and counted; regions with a zero variance estimate (13, 39, 55 by item) receive no shrinkage, which is conservative.

6.3 Survey documentation

Population and selection. The European Social Survey samples residents aged 15 and over in private households, regardless of nationality or citizenship, by strict probability methods in each participating country; designs are country-specific and include stratified and multi-stage cluster selection. We use the integrated data files for rounds 9, 10 and 11 with their design variables (stratum, psu, region, inclusion probability prob) and the design weight dweight. The analysis unit here is the region within a country-round, defined by the file’s region variable.

Fieldwork dates and response rates. These are country-specific, published by round, and **must be tabulated in an appendix before submission**; they are not reproduced here.

Exact question wording. Three items, fixed in advance. Quoted from the ESS source questionnaire; the authors should confirm the wording against the round 9–11 source questionnaires, since only the round 1 instrument was consulted.

trstpr1 (B7). “Using this card, please tell me on a score of 0–10 how much you personally trust each of the institutions I read out. 0 means you do not trust an institution at all, and 10 means you have complete trust. Firstly ... [country]’s parliament?”

stflife (B29). “All things considered, how satisfied are you with your life as a whole nowadays? Please answer using this card, where 0 means extremely dissatisfied and 10 means extremely satisfied.”

happy (C1). “Taking all things together, how happy would you say you are? Please use this card,” with endpoints labelled extremely unhappy and extremely happy.

Estimator and variance estimator. For region r of country-round c the direct estimate is the design-weighted proportion $\hat{F}_{cr} = \sum_{i \in r} w_i z_i / \sum_{i \in r} w_i$ with $z_i = \mathbf{1}\{\text{response} \leq t_0\}$, and its design variance is estimated by ultimate-cluster Taylor linearisation over primary sampling units within strata, in the form given in Section 6 below. Degrees of freedom are counted as clusters minus strata within the country-round.

6.4 The score and its variance must match

The score is $Y_{cr} = \hat{F}_{cr} - \hat{F}_c$, a deviation from an *estimated* country centre. Its design variance is not that of $\hat{F}_{cr}$: it is that of the difference, and we compute it from the Taylor linearisation

$$u_i = \frac{w_i \mathbf{1}\{i \in r\} (z_i - \hat{F}_{cr})}{W_r} - \frac{w_i (z_i - \hat{F}_c)}{W_c}$$

aggregated to PSU totals over the whole country-round. **This is not a technicality.** With the wrong variance the fitted structure exponent has median 0.53 and its confidence set contains the elementary value $\gamma = 1$ ($D \propto 1/n$) in *none* of 45 configurations, including one implausible negative estimate; with the right one the median is 1.14 and the set contains $\gamma = 1$ in 17 of 45.

6.5 What the arms do on the same data

comparison	range	narrower, and median
proposed / normal-theory comparator	0.729–1.014	44 of 45 , 0.821
proposed / structure-free	0.459–1.034	43 of 45, 0.830
proposed / uncorrected	0.422–0.867	45 of 45, 0.730

The comparison against a comparator sharing the target reproduces the simulation on real data—median 0.821 against a simulated 13.9–22.2% reduction—and is the robust one, because both arms are built on the same variance. Realised design shares are $\hat{\rho}^2 \in [0.22, 0.61]$: the regime the correction is for.

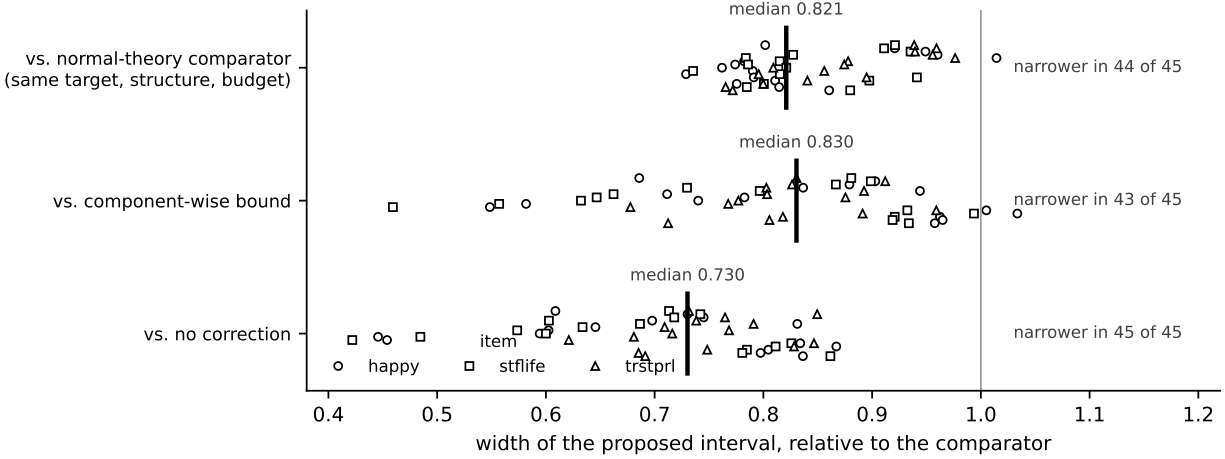


Figure 2: Width of the proposed interval relative to each comparator, over the 45 item-by-round-by-country configurations of European Social Survey rounds 9–11. Bars are medians. All three comparisons use the variance of the centred score (Section 6); with the uncorrected variance the middle comparison moves to a median of 0.992. **No coverage is claimed:** the latent target is unobserved.

Alt text: Three horizontal strips of points, one per comparison, showing width ratios for 45 configurations. Almost all points lie to the left of a vertical line at one, meaning the proposed interval is narrower; median bars are at 0.82, 0.83 and 0.73.

6.6 Which assumptions the data support

The pivot needs normality of the standardised regional deviations. Skewness and excess kurtosis are -0.47 to 0.62 and 0.3 to 4.4 for the first item, and reach 2.30 and 12.6 , and 2.81 and 17.6 , for the other two. **The condition is reasonably supported for one item of the three and clearly not for the others**, and we report that with their results.

The structure model explains 0.09 to 0.23 of $\log \hat{D}$ once the right variance is used—moderately at best.

This is not the quantity that separated the two columns of Section 5, and the two must not be compared. There the structure R^2 was computed against the *true* design variances, which a simulation supplies and a survey does not; here it is computed against estimated ones, and estimation noise in $\log \hat{D}$ depresses it by an amount we cannot remove. A reader should therefore not conclude from 0.09 – 0.23 that these data sit in the regime where the procedure lost containment—

nor that they sit in the safe one. The honest statement is that the diagnostic available on real data does not identify which column a real survey belongs to, and supplying that mapping would require knowing the design variances one is trying to estimate. An analyst can compute that quantity, and across these 45 configurations it is associated with whether using the structure pays (correlation -0.30). **We report this as an observable exploratory diagnostic and not as a criterion:** both quantities are built from the same variance estimates so part of any association is arithmetic, and a good structure fit does not imply coverage, which is not measured here.

6.7 The approximation this analysis rests on

Correcting the variance does not restore exchangeability. A common estimated centre makes the scores dependent, and we measure it: the mean within-country correlation among distinct regions is -0.232 , -0.116 , -0.046 , -0.029 for country-rounds with at most 5, 6–10, 11–20 and more than 20 regions—almost exactly $-1/(n_r - 1)$, the classical dependence of deviations from a common mean. It is negative, hence the benign direction for a maximum-based rank statistic, and it shrinks with the number of regions, but it is not zero.

Status of this analysis. An application of the procedure under a stated approximation: the centre is estimated rather than known, and the scores carry a measured dependence of about $-1/(n_r - 1)$. **The guarantee of Section 4 is not attached to it.**

7 Discussion

7.1 Why a reader would use this, and on what evidence

The question a reader brings is why to use this rather than what they already have. The answer is in three parts and each has a different kind of support behind it.

Because the quantity you must pin down is not the one you think. The interval depends on the two scales only through their ratio, and a requirement stated on the variance component overstates

what the band needs by ρ^{-4} in the number of populations. This is algebra, verified numerically, and it applies whatever construction is used. It is also, by (6), a property of the problem: the same $\rho^4/\{\varepsilon^2(1 - \rho^2)^2\}$ appears in a two-point lower bound on *any* procedure that certifies the ratio, within a factor of about 2.5 of what our estimator achieves.

Because how an interval reacts to scale uncertainty is a property of its form, and forms differ.

Proposition 1 is exact, and it says the deconvolution form pays $\lambda^{\rho^2/2}$ where every construction proportional to a square-root scale bound pays $\lambda^{1/2}$. That covers a class—normal-theory intervals, EBLUP prediction intervals for a target with a new random effect, parametric bootstrap intervals built on them—rather than one comparator. **It does not rank them absolutely:** the oracle width ratio differs between constructions and the elasticity does not supply it. What it supplies is the part that reverses, and the reversals are what a practitioner would otherwise experience as unpredictability.

Because the resulting procedure is computable and its cost is measured. Under Assumption 1 it has the guarantee (10), the supremum is bounded rather than searched, and against a comparator sharing the target, information and level it is 14 to 22% narrower in simulation and narrower in 44 of 45 real survey configurations.

7.2 What is not established

The gap between the theorem and the application is the paper’s main weakness, and naming it does not close it. Assumption 1 is a normal model with a known centre and independent populations. In the European Social Survey the centre is estimated, the scores carry a measured dependence of about $-1/(n_r - 1)$, the standardised deviations are badly non-normal for two of three pre-specified items, and the χ^2 law for a design-variance estimator is not established for complex designs (McGovern, Fuglstad, and Wakefield, 2026). We report the ESS analysis as an application under those approximations with the guarantee not attached, and a reader who needs a guarantee on survey data does not get one here.

Given the same structure information, a component-wise construction is close. Against a component-wise bound that estimates the same structure and spends the same budget, the ratio construction is 0.7 to 2.3% narrower. Most of the gain over a structure-free baseline is the use of a variance structure, not the ratio formulation. We prefer the ratio form because it has one fewer budget piece, a shorter chain of sufficient conditions, and—among the constructions we implemented—a conservative computational bound over a continuous structure confidence set. That no such bound has been built for the component-wise form is not evidence that none exists.

We do not benchmark against the conformal construction that avoids the problem. Zhang and Tuoto (2026) obtain conditional validity for in-sample area expectations without estimating the sampling variances, and their reported half-widths grow with the design share where a direct interval’s do not. We have not run their construction against ours, because the two cover different objects and a width comparison across targets is not informative. A reader whose target is in-sample should read that comparison as *unmade*, not as *decided* here.

The lower bound is about certifying the ratio, not about width. Equation (6) says no procedure can pin κ down with fewer degrees of freedom. It does *not* say no interval can be narrower: an interval may be valid without certifying κ , by being conservative in some other way, and we have not proved a width lower bound. Nothing here establishes that the proposed interval is optimal. In the equal-variance submodel the inferential problem is the classical one for an intraclass correlation, where both the exact interval and the design trade-off are known; what the bound adds is the assurance that (5) is not an artefact of the estimator we happened to write down.

The gain is conditional on the structure family. A confidence set for a structure parameter is not protection against the family being wrong. Where the family fails, containment fails while the final coverage does not, and the latter is downstream slack rather than a guarantee.

And what the registered test refuted. Equation (8)’s invariance in the design share is asymptotic, not exact, and its constant needed a $1 + 1.9/K$ correction whose value we fit rather than derive. The order-statistic expansion would supply it; we have not done that.

7.3 Scope

The target here is a scalar population summary. We do not place a whole-distribution simultaneous band and the scalar result under one guarantee: extending Proposition 3 to coordinate-varying scales requires a simultaneous bound on a vector of correction factors, with a multiplicity cost we have not paid.

7.4 What would settle the open questions

The dependence induced by an estimated centre is the first. It is measured, its form is classical, and a construction restoring exchangeability under it would convert the ESS analysis from an approximation into an application of the theorem. The second is the structure family: the diagnostic reported in Section 6 is exploratory and its association with the payoff needs a survey system not used to find it.

Data availability

No microdata are redistributed with this work. The European Social Survey integrated data files for rounds 9, 10 and 11 are obtained from the ESS data portal under its End User Licence; the replication materials include a script that retrieves them through the portal’s application programming interface, so the analysis reproduces from source for any licence holder. No user identifier is stored in the replication materials and the local cache is written outside them. All simulation results reproduce from the replication materials alone. Those materials are supplied with this submission for the reviewers.

Replication

Every numerical statement in this paper corresponds to an entry in a claims file in the replication materials, which also contains the protocol written before each experiment was executed, including its predictions, and records the claims that were subsequently retracted.

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